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Ivar Cashin Eriksson

Dear Professor Rasmussen and Members of the Selection Committee,

I am applying for the PhD in Causal Machine Learning for Real World Evidence data. After seven years working with machine learning in research and industry, I want to return to research and spend more time developing methods and investigating the questions behind them. This project interests me because it connects statistical methodology with understanding disease progression and treatment response. It would allow me to build on my experience in medical machine learning and the causality and computational statistics courses I took at ETH Zürich.

In my previous work at RaySearch Laboratories, I researched methods for automated radiotherapy planning, first through my master's thesis and then as a full-time researcher. I used autoencoders to represent patient anatomy and sparse Gaussian processes to predict radiation dose distributions. I also worked on optimisation methods to better reflect clinical priorities. Working with medical physicists and clinicians taught me to examine how modelling choices relate to the decisions a system needs to support. I enjoyed the combination of mathematical work, practical constraints, and discussion with specialists.

Since then, I have taken responsibility for larger ML systems. At Valcon, I led projects from problem formulation through deployment, including a real-time neural-network pipeline combining text and categorical data. At Signum Life Science, I now lead the architecture and development of a forecasting platform supporting thousands of independently trained models. This involves data validation, reproducible evaluation, experiment tracking, and batch inference. These roles have taught me to structure complex software, investigate failures, and make technical decisions together with colleagues and domain experts.

I also enjoy building things independently. My personal project, [Iris](#), combines computer vision and similarity search to make products within webshop images searchable. I built the workflow from object detection and image embeddings through to backend APIs and a browser extension. It gives me room to explore unfamiliar methods and tools, and to work through the details of turning an idea into a complete application.

My degree in Engineering Physics, specialising in Statistical Learning and Data Analytics, provides a strong mathematical foundation for this work. I would bring experience developing and evaluating machine-learning methods, building reproducible workflows in Python, and working with clinical specialists. In the Rasmussen Group, I would like to build on this experience and deepen my understanding of causal inference and its application to real world evidence.

I welcome the opportunity to discuss the project and how my background could contribute to the research.

Kind regards,
Ivar Cashin Eriksson